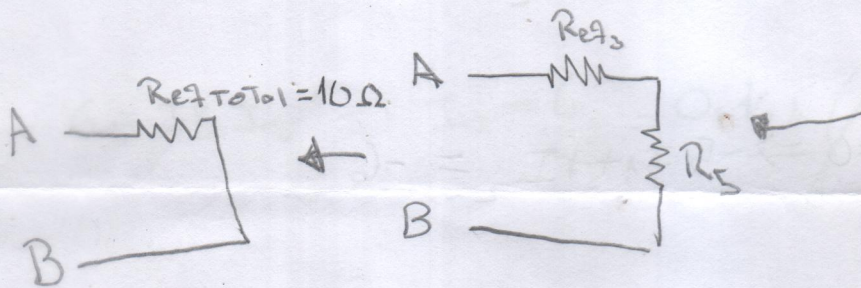
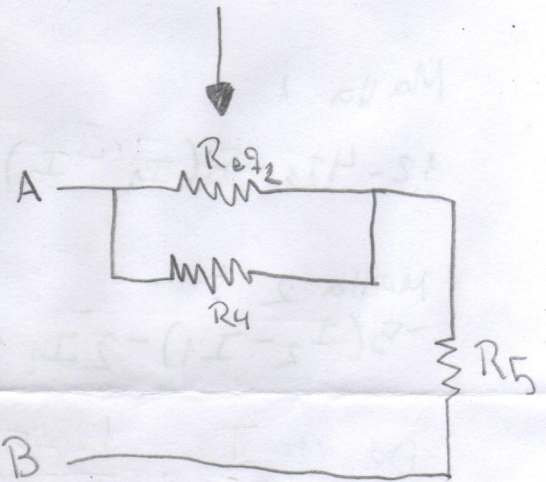
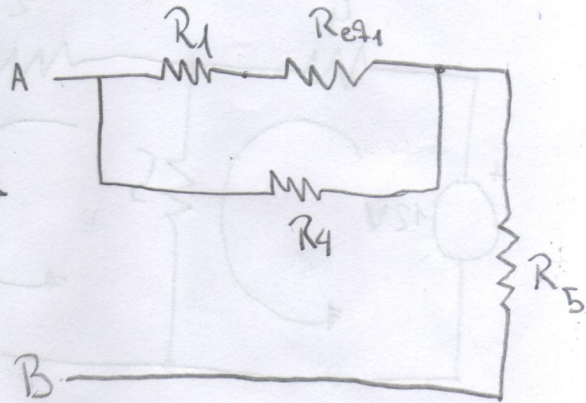
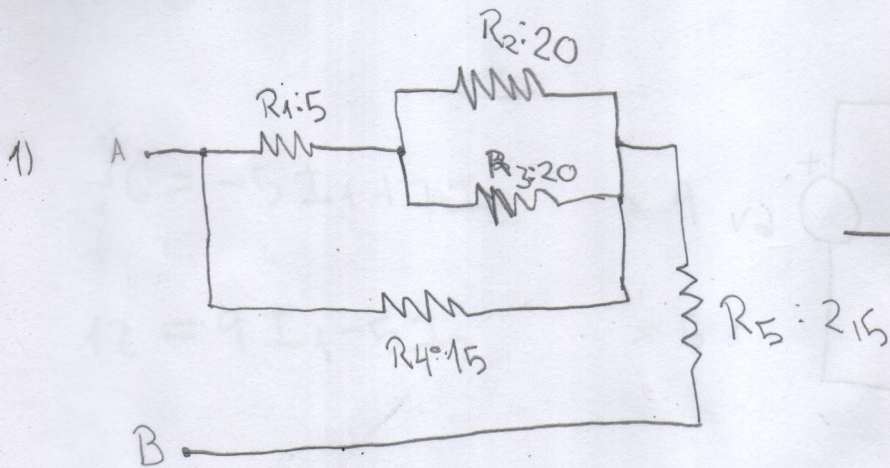


Comanda 15/05/26
Br: Tirzo Mata



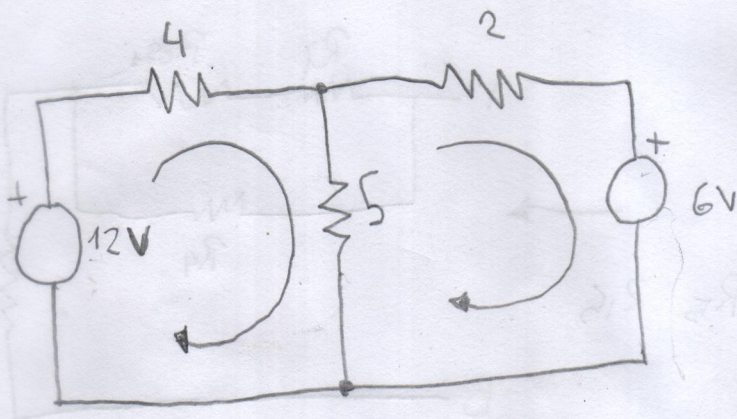
$$\frac{1}{R_{eq}} = \frac{1}{20} + \frac{1}{20} = \frac{40}{400} \Rightarrow R_{eq} = \frac{400}{40} = 10 \Omega$$

$$R_{eq2} = 5 + 10 = 15 \Omega$$

$$\frac{1}{R_{eq3}} = \frac{1}{15} + \frac{1}{15} = \frac{30}{225} \Rightarrow R_{eq3} = \frac{225}{30} = 7.5 \Omega$$

$$R_{eqTotal} = 7.5 + 2.5 = 10 \Omega$$

3) Analisis de mallas



Malla 1

$$12 - 4I_1 - 5(I_1 - I_2) = 0 \Rightarrow 9I_1 - 5I_2 = 12$$

Malla 2

$$-5(I_2 - I_1) - 2I_2 - 6 = 0 \Rightarrow -5I_1 + 7I_2 = -6$$

$$\text{De (1)} \quad I_1 = \frac{12 + 5I_2}{9}$$

$$\text{En (2):} \quad -5\left(\frac{12 + 5I_2}{9}\right) + 7I_2 = -6$$

$$I_2 \approx 0,16A \quad I_1 \approx 1,42A$$

$$I_R \approx 1,26A$$

$$12 - 4 - 5 = 0$$

$$-6 = -5I_1 + 7I_2 \quad \times 9$$

$$12 = 9I_1 - 5I_2 \quad \times 5$$

$$-54 = -45I_1 + 63I_2$$

$$60 = 45I_1 - 25I_2$$

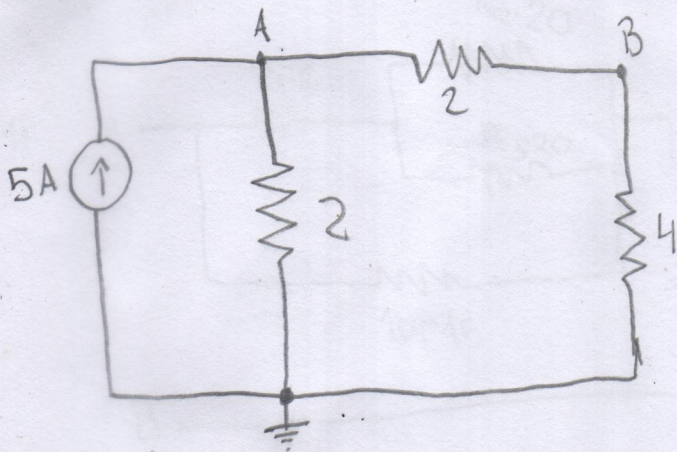
$$6 = 38I_2 \Rightarrow I_2 = \frac{6}{38} = 0,16A$$

$$\frac{6 + 7(0,16)}{5} = I_1 = 1,42$$

En el resistor 5

$$I = 1,42 - 0,16 = 1,26A$$

2) Analisis Nodal



En el Nudo A

$$5 = \frac{V_A}{2} + \frac{V_A - V_B}{2} \Rightarrow 10 = 2V_A - V_B$$

En B

$$\frac{V_A - V_B}{2} = \frac{V_B}{4} \Rightarrow 2V_A - 2V_B = V_B \Rightarrow 2V_A = 3V_B$$

Sustituyendo

$$3V_B - V_B = 10 \Rightarrow 2V_B = 10 \Rightarrow V_B = 5V$$

$$2V_A = 3(5) \Rightarrow V_A = 7.5V$$